

Critical slowing down

- Near criticality, long-distance modes evolve slowly.
- Global flows must learn both infrared and ultraviolet structure.
- Residual local errors accumulate with volume, typically suppressing global Metropolis acceptance as $P_{\text{acc}} \sim e^{-cV}$.

The scalar ϕ^4 theory

Action:

$$S_f[\phi] = -2\kappa \sum_{x,\mu} \phi_x \phi_{x+\hat{\mu}} + \sum_x [\phi_x^2 + \lambda(\phi_x^2 - 1)^2]$$

In d=2:

$$\lambda = 1.0, \quad \kappa_{\text{cr}} = 0.3401.$$

Binder cumulant:

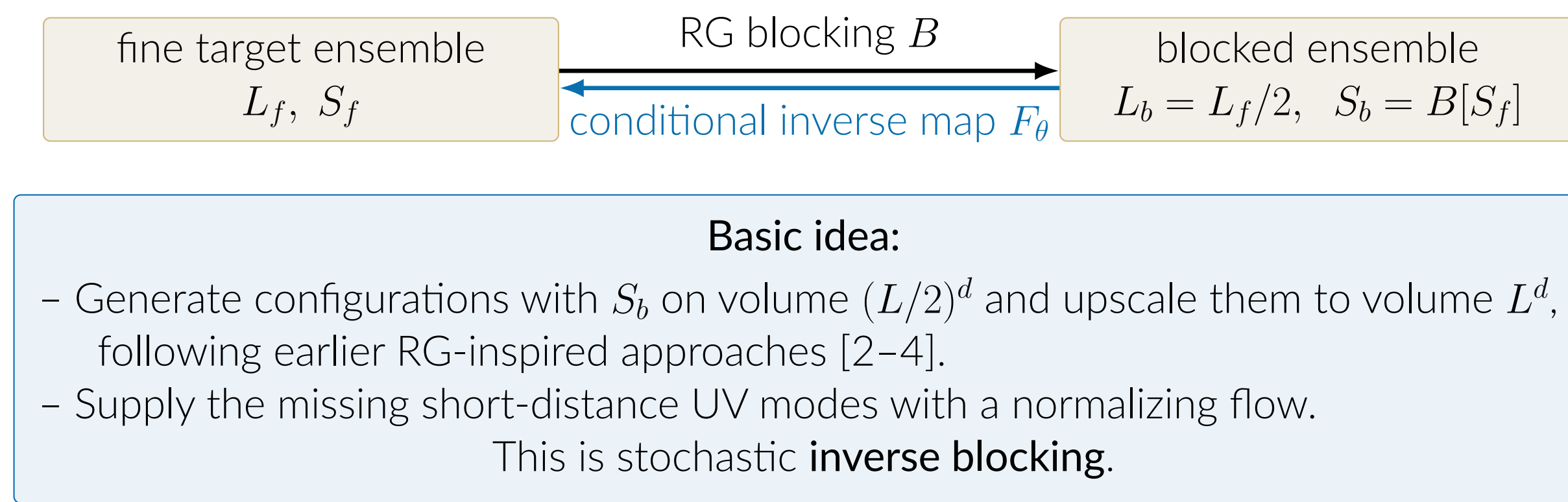
$$U_4(L=16) = 0.6051(16), \quad U_4(L=32) = 0.6080(11),$$

consistent with the 2D Ising value $U_4^* \simeq 0.61069$.

Renormalization Group block transformation

Under a scale-2 RG block transformation

- $S_f \rightarrow S_b = B[S_f]$; S_b is complicated, but corresponds to $L \rightarrow L/2$, $\xi \rightarrow \xi/2$
- the infrared properties are unchanged while coarse graining removes ultraviolet information



Approximate perfect blocking

Perfect Blocking:

Require that the critical action blocks into itself, $\mathbf{S}_b(\kappa_{\text{cr}}) = \mathbf{B}[\mathbf{S}_f(\kappa_{\text{cr}})]$:

- Choose a translationally invariant finite-range kernel K
- Form $\psi = K\phi$ and retain the even-even sites ψ_{ee}
- Compare ψ_{ee} with directly generated coarse fields ϕ_c
- If $S_c = B[S_f]$, all operator expectation values of ψ_{ee} and ϕ_c must agree
- Optimize K using a covariance-weighted mismatch of local observables

A five-parameter 5×5 kernel already gives close coarse-distribution matching.

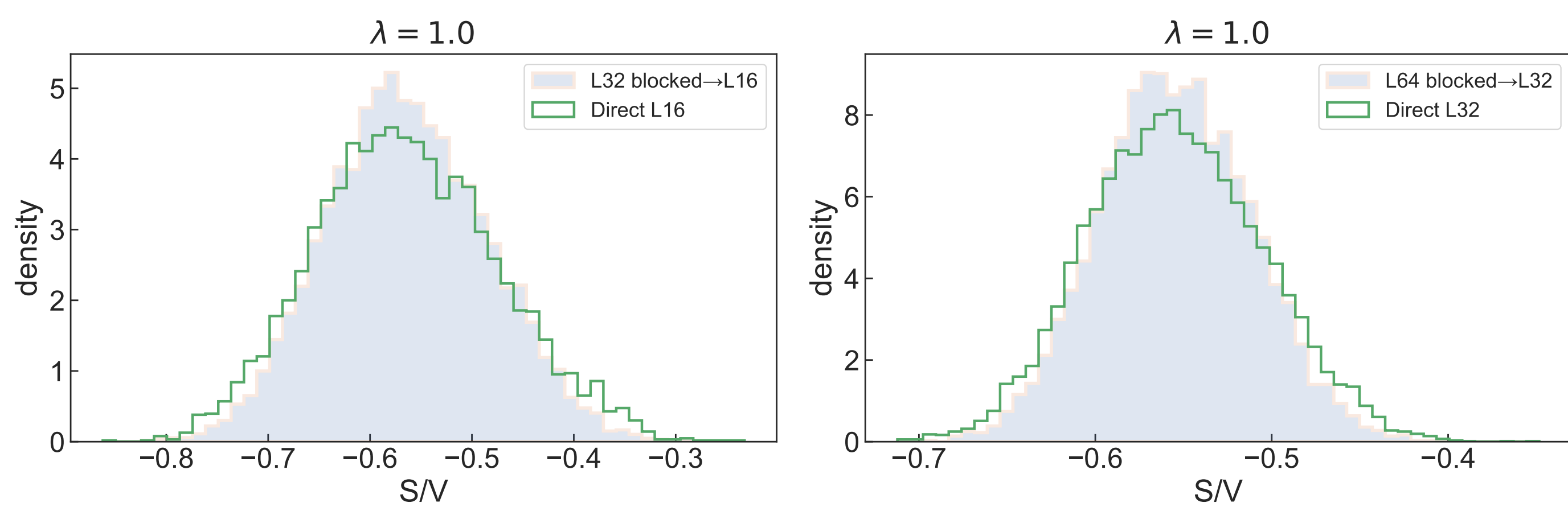


Figure 1. Action-density distributions for directly generated and blocked ensembles. Left: $32 \rightarrow 16$. Right: $64 \rightarrow 32$.

What “perfect” means here

- The goal is practical, not formal perfection: blocked and direct coarse ensembles must overlap well enough to preserve the infrared modes
- Any remaining mismatch is corrected by the exact accept/reject step.

Conditional upscaling flow

The fine proposal is parameterized by a coarse field and detail variables:

$$\tilde{\phi}_f = K^{-1}F_\theta(\phi_c, z), \quad z : \text{Gaussian random noise}$$

Optimize the initial inverse map $F_\theta(\phi_c, z)$ in stages:

- Embed ϕ_c , generated with $S(\kappa_{\text{cr}})$, on the even-even sites
- Generate edge sites (1,0) and (0,1) first, then body-centered sites (1,1) [5]
- Apply the invertible kernel K^{-1}
- Train by matching local observables of native and proposed fine ensembles
- Add tail-sensitive loss terms to reduce rare large deviations

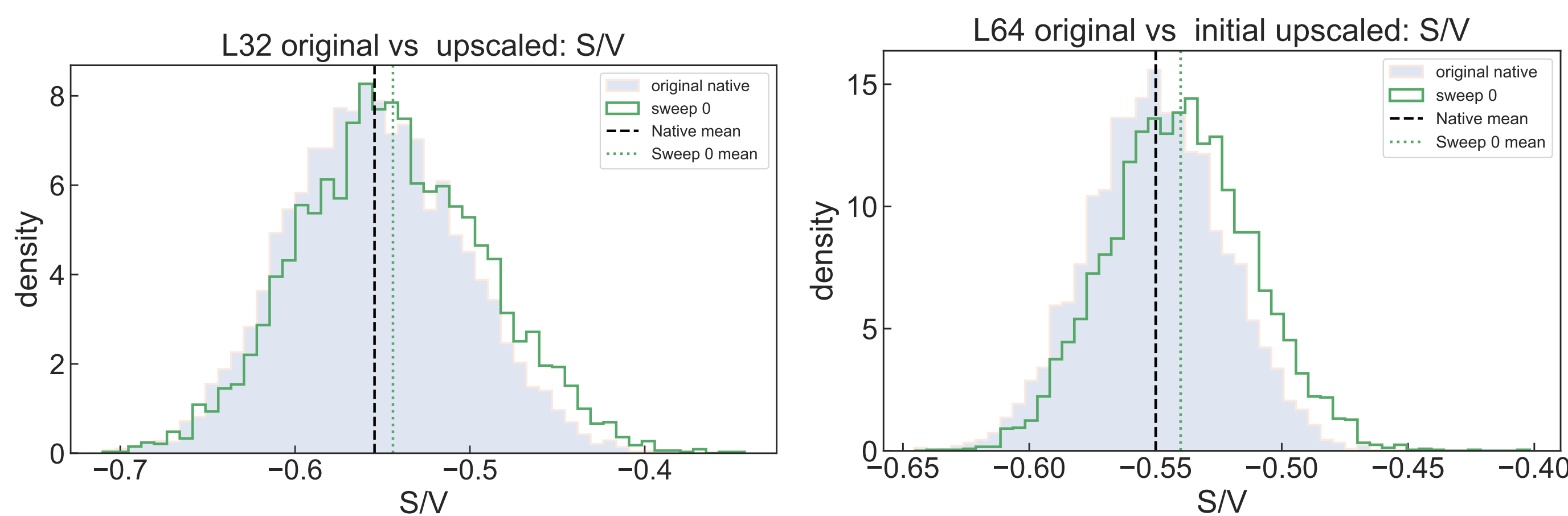


Figure 2. Native and initially upscaled action-density distributions. Left: $16 \rightarrow 32$. Right: $32 \rightarrow 64$.

- The initial proposal is slightly displaced but covers the target support
- Global whole-volume correction is inefficient: acceptance $\lesssim 3\%$ on $16 \rightarrow 32$
- Instead: exact local patch updates repair the remaining mismatch

- The blocking transformation is local, conditional flow is not.
- A flow trained only on $L = 16 \rightarrow 32$ should be retrained for $L = 32 \rightarrow 64$. (Not yet done.)

Why inverse blocking?

- Generate equilibrated coarse configurations from an RG-blocked ensemble.
- Use a compact conditional flow to reconstruct only the missing ultraviolet modes.
- Remove residual bias with exact patchwise coarse and detail updates.

For fixed patch size, the work per sweep scales as $\mathcal{O}(V)$.

Flow initialization • exact coarse • detail updates

- **Initialize once:** draw $c \sim e^{-S_c}$ and use the upscaling flow to generate d_{01}, d_{10}, d_{11} .

- **Reconstruct:**

$$(c, d_{01}, d_{10}, d_{11}) \rightarrow \psi \rightarrow \phi = K^{-1}\psi$$

Subsequent updates act directly on the coarse and detail fields.

- **Coarse update:** propose $c \rightarrow c'$ with MH under S_c and accept with $\log A_c = -S_f(\phi') + S_f(\phi) + S_c(c') - S_c(c)$.
- **Detail updates:** update d_{01}, d_{10}, d_{11} sequentially with symmetric proposals and $\log A_d = -S_f(\phi') + S_f(\phi)$.
- **Subvolume schedule:** on larger lattices, update one residue class at a time; one recorded sweep visits all residue classes.

The flow determines only sweep 0; all subsequent evolution is exact Metropolis–Hastings targeting S_f .

Rapid re-thermalization

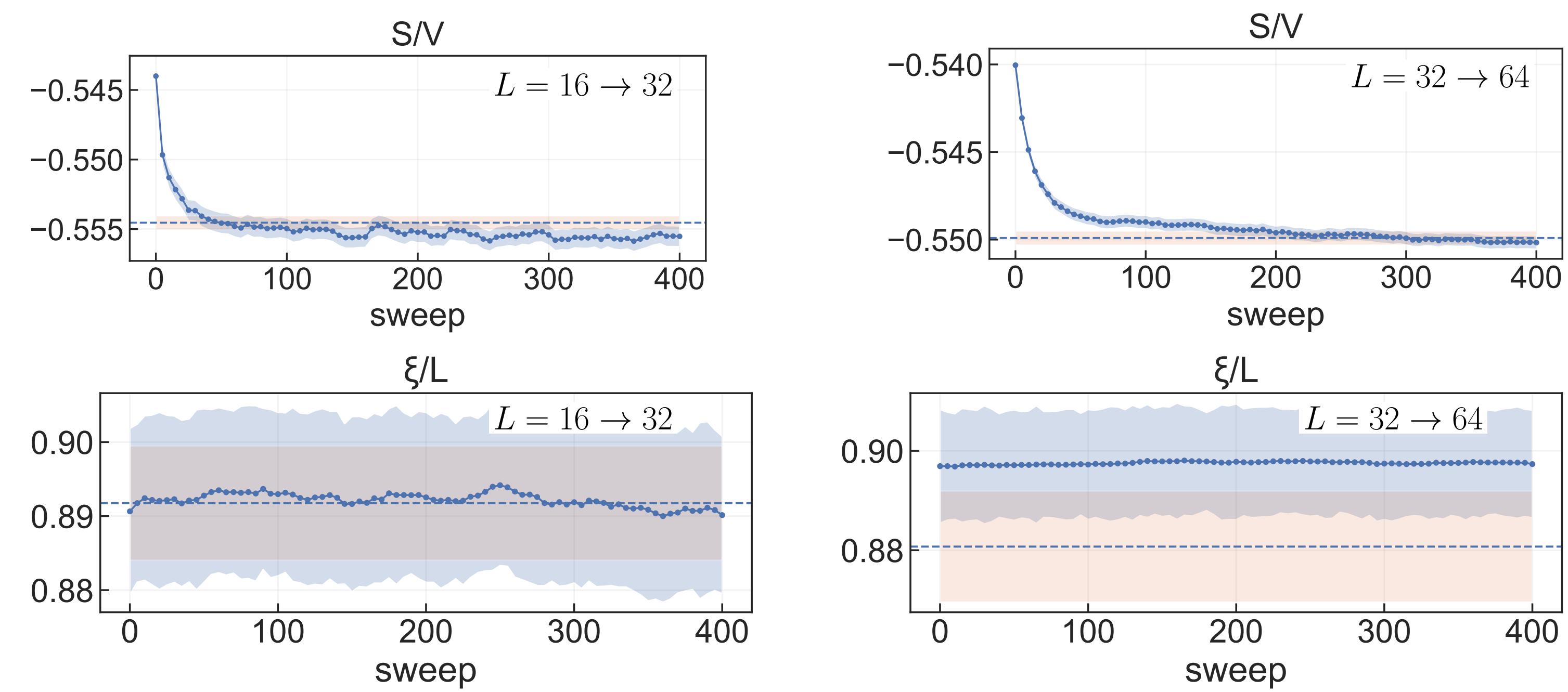


Figure 3. Re-thermalization of action-density and ξ/L . Left: $16 \rightarrow 32$, Right: $32 \rightarrow 64$

Most observables agree with the native ensemble by sweep 200; the sweep-400 distributions show no visible residual bias.

The action-density distribution rapidly converges to the native target.

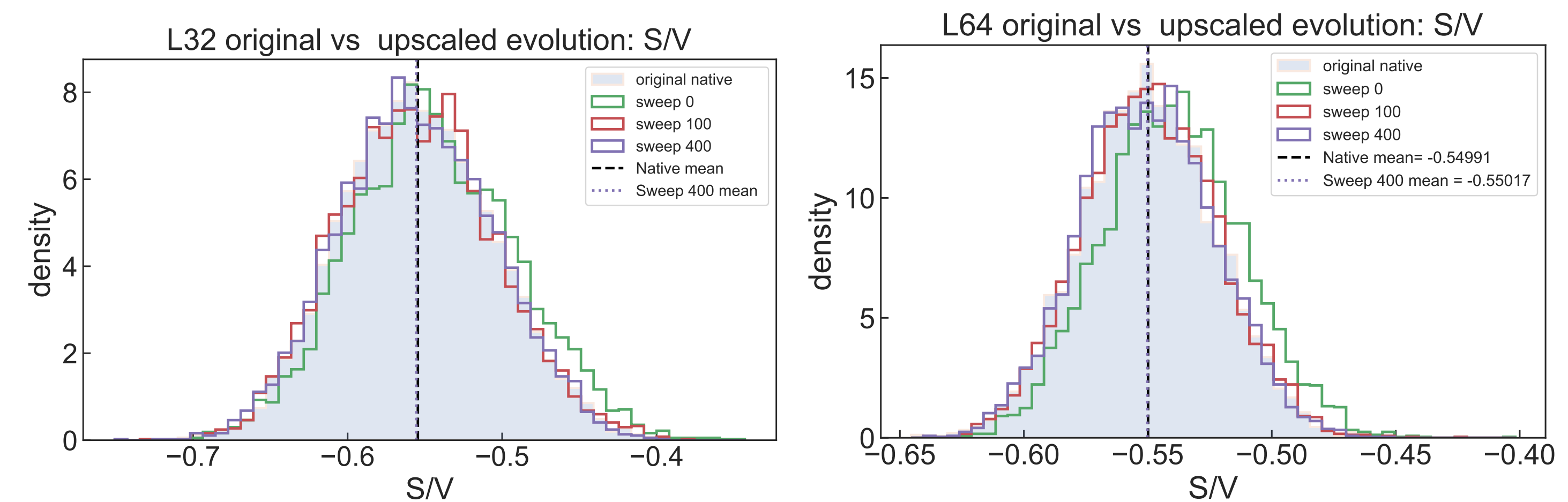


Figure 4. Rethermalization of the action-density distribution after 0, 100, and 400 sweeps.

Proposed update steps:

- Generate coarse configurations on $(L/2)^d$
- Upscale to L^d with the conditional flow
- Rethermalize with exact coarse and detail patch updates
- Retain one configuration per chain at a fixed sweep

Summary

- Approximate perfect blocking transfers equilibrated infrared modes from coarse to fine lattices
- A compact conditional flow initializes the missing ultraviolet degrees of freedom
- Exact coarse-corrected and sequential detail MH updates remove the initialization bias.
- The blocking kernel transfers directly to larger volumes; improving flow portability remains future work
- The cost per sweep scales as $\mathcal{O}(V)$; the remaining question is the scaling of the rethermalization time

Improving the initial flow should reduce the required number of rethermalization sweeps.

For complementary results, see the related talk by L. Parato.

References

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- [4] Bauer et al, SciPost Phys. 19 (2025) 3, 077.
- [5] A. Singha et al, arXiv:2604.10209

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